Carbon Dioxide Pipelines

Graeme King, Tran Mah-Paulson, and Nicki Robertson

Contents

G. King
Tensor Engineering Ltd., Calgary, AB, Canada
e-mail: graeme.king@tensorengineering.com

T. Mah-Paulson (✉)
TDMP Consulting, Edmonton, AB, Canada
e-mail: tran@tdmpconsulting.com

N. Robertson
University of Alberta, Edmonton, AB, Canada
e-mail: nicolele@ualberta.ca; nrobertson@pembina.com

ABCM – Brazilian Society of Mechanical Sciences and Engineering et al. (eds.),
Handbook of Pipeline Engineering, https://doi.org/10.1007/978-3-031-05735-9_46-2

Abstract

Carbon dioxide pipelines are similar to natural gas pipelines in many ways. Where they are similar, the pipeline industry has considerable experience and can rely on existing standards and regulations. But where they are different, the pipeline industry has less experience, and designers and operators must rely more heavily on basic engineering principles. This chapter focuses on the differences and the steps required to deal with the differences. In all cases, the differences are driven by the differences in properties between CO_2 and natural gas. The differences include operating pressure ranges, running ductile fracture control, pressure fluctuations and fatigue, unique corrosion challenges, interaction with non-metallic components, blowdown stack design, and emergency response plans in the unlikely event of a line break.

The intent is to provide the reader with a basic understanding of the design, operation, and maintenance of carbon dioxide pipelines where they are different from natural gas pipelines and where existing regulations, codes, and standards do not fully cover the special features of carbon dioxide pipelines. This basic understanding needs to be combined with sound engineering judgement to develop safe and reliable designs and procedures. In some cases where there is

no consensus amongst experts in the field, the authors express their own opinions about how to proceed, but those opinions could change as more experience with carbon dioxide pipelines is obtained.

Keywords

Carbon dioxide · CO_2 · Propagation · Burst testing · Decompression behavior · Dense phase · Gaseous phase · Asphyxiation · Enhanced oil recovery · Storage · Utilization

Introduction

Climate change is driving plans to decrease the amount of anthropogenic carbon dioxide (CO_2) released into Earth's atmosphere by implementing carbon capture, utilization, and storage (CCUS) projects. These projects use pipelines to transport CO_2 captured from high volume emitters (such as, refineries, cement plants, fertilizer plants, hydrogen production, DAC facilities, etc.) to utilization and storage sites. Over the last 50 years, approximately 9000 km of carbon dioxide pipelines with a total capacity of around 200 $MtCO_2$/year have been developed globally, while the IEA Net Zero Scenario for 2030 requires a capacity of 1190 $MtCO_2$/year (IEA 2022), which is six times the existing capacity in one-sixth the time. This large increase in capacity over a short period will require a matching increase in the total length and capacity of carbon dioxide pipelines.

The total length of carbon dioxide pipelines operating in the world today is less than 0.4% of the total length of existing natural gas pipelines (9000 vs 2,500,000 km) (IEA 2022; Peletiri et al. 2018; CIA Factbook 2015). Since the installed length of carbon dioxide pipelines is relatively small and most of them are relatively new, the pipeline industry has considerably less practical experience with carbon dioxide pipelines than with natural gas pipelines, and, in some cases, additional research and development is needed to fill the gaps in the knowledgebase of carbon dioxide pipelines.

Carbon dioxide pipelines are similar in many ways to natural gas pipelines. At the same time, carbon dioxide has some distinctly different properties from natural gas, which require different designs and operational procedures to handle them safely. Where the properties are the same, the pipeline industry has extensive real-world experience that can be relied upon to develop safe cost-effective solutions. But where they are different, less practical experience is available, and pipeline designers and operators must rely on fundamental engineering principles to design and operate safe and reliable pipelines. Table 1 lists areas where the design and operation of carbon dioxide pipelines is similar to natural gas pipelines and existing pipeline codes and standards can be used, and areas where they are different and require special designs and procedures based on fundamental engineering principles for the safe and reliable transportation of CO_2 by pipeline.

Table 1 Comparison of carbon dioxide and natural gas pipelines

Parameter	Similar	Somewhat different	Very different	Significantly different
Phase behavior				X
Effect of impurities				X
Long ductile fractures			X	
Pipe material testing and specs			X	
Fittings material testing and specs	X			
Welding		X		
Construction methods	X			
Non-metallics			X	
Operating pressure				X
Depressurization and blowdown				X
Pipeline maintenance		X		
Lubrication			X	
Fire and explosion risk				X
Asphyxiation risk				X
Public education and awareness			X	
Emergency response			X	
Integrity management		X		
Corrosion		X		
Inhibitors			X	
Deactivation and Abandonment	X			

The differences between the two kinds of pipeline are a direct result of fundamental differences between the properties of the two fluids. A strong understanding of the fundamental differences between the two fluids and the impact the differences have on operating conditions, material selection, compositional limits, operating procedures, stakeholder and public relations, and emergency response plans need to be combined with sound engineering judgement to develop safe and reliable designs and procedures for carbon dioxide pipelines. In specific cases where there are holes in the knowledgebase, safety calls for designers and operators to build fences around the holes or undertake additional testing to fill the holes. In instances where there is little experience, such as loss of containment, and no consensus has been reached about the best way to proceed, the authors express their own opinions, but it should be remembered that those opinions could change as more experience with carbon dioxide pipelines is obtained.

History of Carbon Dioxide Pipelines

Carbon dioxide pipelines have been around since the early 1970s, when the first large carbon dioxide pipeline was built in the USA (Doctor et al. 2005). In 2022, pipelines are used in 30 operational CCUS projects to transport carbon dioxide from industrial and natural sources to either storage reservoirs or enhanced oil recovery (EOR) fields (Global CCS Institute 2022). Table 2 lists some of the larger carbon dioxide pipelines operating in the world today.

Codes, Standards, and Regulations

Pipeline codes, standards, and regulations began to include specific clauses for carbon dioxide service in the mid-1980s. Carbon dioxide pipelines can operate in the gas phase either at low pressure below the phase envelope or at high pressure above the phase envelope. While existing standards provide guidance on a handful of design issues, the differences between dense phase and gas phase carbon dioxide pipelines are still not clearly delineated, and in some cases gas phase CO_2 pipelines are specifically excluded from standards and regulations.

Some of the more established code committees have "squeezed" specific clauses related to carbon dioxide pipelines into existing oil and gas pipeline codes and standards, while other jurisdictions have established codes and standards devoted entirely to carbon dioxide pipelines. In addition, most codes and standards are more goal oriented than prescriptive. Because of these factors, the requirements of codes and standards for carbon dioxide pipelines are not always clear, and it is advisable to investigate each of the parameters listed in Table 1 and apply fundamental engineering principles to them using the latest information available before finalizing designs and operating procedures. Table 3 summarizes what is covered, at least partially, in some of the more widely accepted pipeline standards and regulations.

Nature of Carbon Dioxide

Figure 1 is the phase diagram for pure CO_2 plotted on a pressure-temperature graph. Two key elements of the phase diagram are the triple point at 0.52 MPa and -56.6 °C and the critical point at 7.39 MPa and + 31.1 °C (Wang et al. 2019). At the triple point, pure CO_2 can exist as a solid, liquid, or gas, while at the critical point, separate gas and liquid phases cease to exist.

An important feature of the phase diagram is that the fluid in the region labelled "gas" can be compressed into the "supercritical" region and then cooled into the "liquid" region without changing phase, that is, without crossing into the two-phase region represented by the line between the triple point and the critical point. Another important feature is that CO_2 in the region labelled "liquid" is not a true liquid. It is dense like a liquid and relatively incompressible, but it has no free surface and expands to completely fill the vessel that contains it, just like a gas. Because there is

Table 2 List of larger scale carbon dioxide pipelines globally

Pipeline	Diameter	Length	Year built	Location	Source of CO_2	Purpose
Canada pipelines						
Quest Pipeline	16″	100 km	2015	AB [Canada]	Industrial	CCS
ACTL	16″	240 km	2020	AB [Canada]	Industrial	EOR
Souris Valley Pipeline	14″ / 12″	320 km	1999	SK [Canada] & ND [USA]	Industrial	EOR
Gulf Coast pipelines						
NEJD, Delta, Free State, West Gwinville	20″, 24″, 20″, 18″	701 km	1986, 2009, 2005, 2008	MS & LA [USA]	Natural: Jackson Dome	EOR
Green Pipeline	24″	515 km	2010	TX & LA [USA]	Industrial & Jackson Dome	EOR
Rocky mountain pipelines						
Greencore Pipeline	20″	373 km	2012	WV & MT [USA]	Industrial	EOR
CCA Pipeline	16″	168 km	2021	MT & ND [USA]	Industrial	EOR
Beaver Creek	8″	74 km	2008	WV [USA]	Industrial	EOR
Permian basin pipelines						
Bravo Pipeline	20″	351 km	1980	NM & TX [USA]	Natural dome: Bravo	EOR
Central Basin Pipeline	26″ / 16″	286 km	1986	TX [USA]	Natural sources	EOR
Cortez Pipeline	30″	803 km	1982	CO, NM & TX [USA]	Natural domes: McElmo / Doe Canyon	EOR
Canyon Reef Carriers	16″	354 km	1972	TX [USA]	Natural sources	EOR
Sheep Mountain	24″	657 km	1983	CO [USA]	Natural sources	EOR
Global pipeline projects						
Snøhvit	8″	153 km	2008	Snøhvit [Norway] Subsea	Gas production	CCS
Uthmaniyah	–	85 km	2015	[Saudi Arabia]	Industrial	EOR
Ordos	–	201 km	2011	Inner Mongolia [China]	Industrial	CCS/ EOR
Jilin	–	20 km	2008	Songyuan [China]	Industrial	EOR

Table 3 Specific requirements for carbon dioxide pipelines in global standards and regulations (as of mid-2023)

Pipeline design codes & standards	Includes specific requirements for carbon dioxide pipelines[a]					
	Substance type	Fracture control	Welding & materials	Operations	Consequence management	Emergency management
CFR 195 Transportation of Hazardous Liquids by Pipelines	Dense phase	✓	✓	–	–	–
CSA Z662 Oil and Gas Pipeline Systems (CSA Group 2019)	Gas & dense phase	✓	✓	✓	✓	✓
ISO 27913 Carbon dioxide capture, transportation and geological storage – Pipeline transportation systems	Gas & dense phase	✓	✓	✓	✓	✓
DNV-RP-F104 Design & Operation of carbon dioxide pipelines	Dense phase	✓	✓	✓	✓	✓
ASME B31.4 Pipeline transportation for liquids and slurries	Dense phase	No	✓	No	No	✓

[a] A check mark in any category indicates that some specific requirements are covered in that category but does not indicate that all requirements are covered

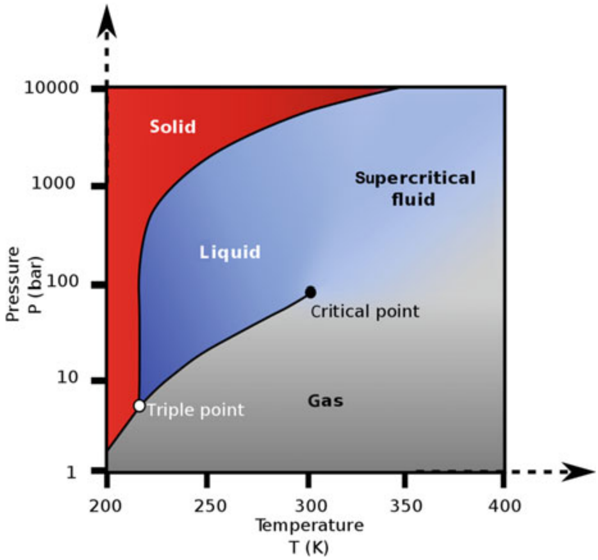

Fig. 1 Phase diagram for pure CO_2

no sharp distinction between "gas" and "liquid" in the region above the phase boundary and because the region is both subcritical and supercritical, it is often referred to simply as the "dense phase," a phrase that was first coined in the early 1970s (King and Katz 1973) to refer to the region above the phase boundary.

If the pressure of the CO_2 is above the critical pressure and its temperature is above the critical temperature, the CO_2 is a true "supercritical fluid." But if the CO_2 is in the region above the phase boundary (the line between the triple point and the critical point) and below either the critical pressure and/or the critical temperature, as it is in most dense phase carbon dioxide pipelines, the CO_2 is neither a supercritical fluid nor a true liquid and is more correctly referred to as a dense phase fluid. In practice, the terms "supercritical," "dense phase," and "liquid" are used interchangeably, but the distinctions between the terms "liquid," "dense phase," "supercritical," and "gas" are useful when designing carbon dioxide pipelines and will be discussed at greater length in the next section when considering the phase behavior of CO_2 with impurities.

Terminology

Understanding terms that are used to describe the phase behavior and thermodynamic properties of CO_2 with impurities is useful, particularly in the region immediately above the phase boundary where fluids have properties that differ significantly from the more familiar properties of ideal gases and liquids. Table 4 defines many of the terms that are used including supercritical, subcritical,

Table 4 Definitions and terminology relevant to carbon dioxide pipelines

Term	Definition
Supercritical pressure	The state where pressure is above the critical pressure
Supercritical temperature	The state where temperature is above critical temperature
Subcritical pressure	The state where pressure is below the critical pressure
Subcritical temperature	The state where temperature is below the critical temperature
Critical point	For a pure substance, the pressure and temperature above which a second phase cannot form
Pseudocritical point	The calculated molal average of the critical points of the components of the mixture
Liquid phase	The phase that acts like an ideal liquid and is relatively incompressible
Gas phase	The phase that generally follows the ideal gas law
Dense phase	The phase that is above the two-phase region between the gas phase and liquid phase
Single phase	Refers to the gas phase, liquid phase, or dense phase outside the two-phase region
Second phase	Refers to either the gas phase or the liquid phase within the two-phase region
Saturation line	The line of equilibrium coexistence of liquid and vapor phases for a pure substance. Triple point and critical point serve as a start and end points, respectively (IGI Global 2022)
Bubblepoint line	The temperature at subcritical pressures where the first bubble of vapor is formed when heating a liquid consisting of two or more components (Wikipedia 2017)
Dewpoint line	The temperature at subcritical pressures where the first drop of liquid is formed when cooling a gas consisting of two or more components (Wikipedia 2022)
Cricondenbar	The critical condensation pressure is the highest pressure beyond which a second phase can no longer form at any temperature
Cricondentherm	The critical condensation temperature is the highest temperature beyond which a second phase cannot form at any pressure

pseudocritical, liquid phase, gas phase, dense phase, second phase, saturation line, bubblepoint line, dewpoint line, cricondenbar, and cricondentherm.

Figure 2 is a pressure-temperature diagram for a mixture of 95% CO_2 with 5% impurities, and it illustrates many of the terms defined in Table 4. One difference between Figs. 1 and 2 is that the two-phase region is no longer represented by a one-dimensional line but has expanded into an observable two-dimensional region.

Figure 2 shows how CO_2 can transition smoothly without any abrupt change in properties from Point A in the gas phase (at about 1 MPa and 5 °C) to Point B near the cricondenbar and cricondentherm (at about 9 MPa and 30 °C), to Point C in the dense phase (at about 14 MPa and 5 °C), and to Point D near the so-called "liquid" phase without passing through the two-phase region and without changing phase.

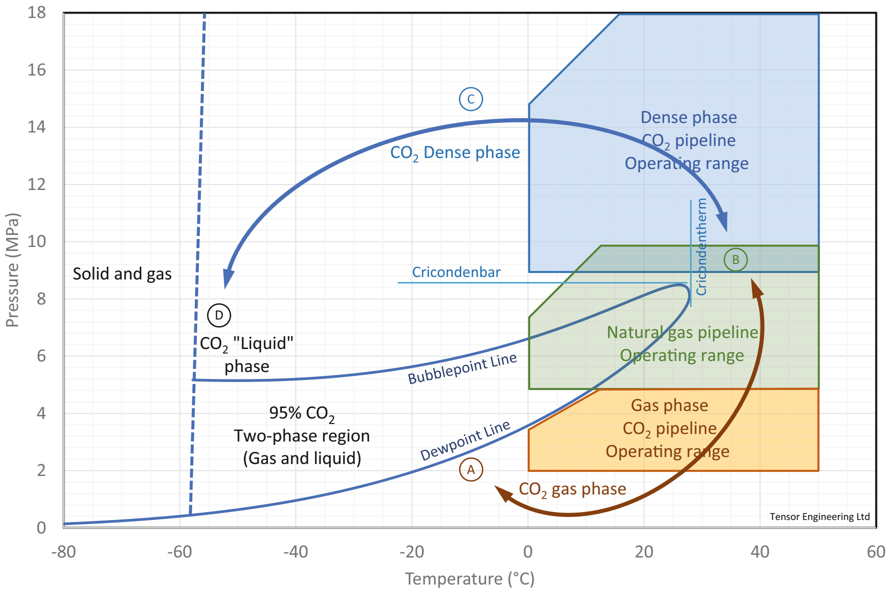

Fig. 2 Phase behavior of CO_2 with 5% impurities

For this reason, the entire region outside the two-phase envelope is referred to as the Single Phase. It is worth emphasizing that CO_2 in the part of the single-phase region labelled "Liquid Phase" is not a true liquid because it has no free surface and entirely fills the containing vessel, just like gas, and if the pressure is released, it does not remain liquid, but evolves into gas and solids.

The term "supercritical carbon dioxide pipeline" is often used to describe carbon dioxide pipelines that operate in the dense phase, but the term "supercritical" can be misleading for several reasons:

1. The CO_2 carried by carbon dioxide pipelines is usually mixed with impurities and does not have a physically unique critical point. Instead, CO_2 mixtures have three closely related terms. They have a pseudocritical point, which is a calculated value that is typically located inside the two-phase region, and they have a cricondenbar and a cricondentherm. The cricondenbar of a mixture is the highest pressure of the phase boundary and is analogous to the critical pressure of a pure substance. The cricondentherm is the highest temperature of the phase boundary and is analogous to critical temperature. But, unlike the critical pressure and critical temperature of a pure substance, the cricondenbar occurs at a temperature different from the cricondentherm, and the cricondentherm occurs at a pressure different from the cricondenbar.
2. Carbon dioxide pipelines operating in the dense phase above the phase boundary are not supercritical along their full length. They usually operate at subcritical

pressures and/or temperatures for much of their length, most noticeably at the suction of compressor/pump stations.
3. The term "supercritical" is not a distinguishing feature for design purposes because nothing noteworthy happens to the properties of CO_2 mixtures in the single phase when passing from subcritical to supercritical pressure or temperature. There is only a smooth and gradual change of properties like anywhere else in the single phase.

The term "supercritical" therefore does not apply to most carbon dioxide pipelines. The term "dense phase" was coined to provide a less misleading term than "supercritical" to describe pipelines that operate at high pressure in the single phase above the two-phase region.

The term "saturation pressure" is often used to describe the pressure along the upper boundary of the phase envelope of CO_2 with impurities. But the term "saturation pressure" only applies to pure substances where the bubblepoint and dewpoint lines are superimposed when plotted on a pressure-temperature diagram. Carbon dioxide pipelines normally transport CO_2 with impurities that have separate bubblepoint and dewpoint lines making the term "saturation pressure" ambiguous. For carbon dioxide pipelines, it is better to refer directly to either the "bubblepoint pressure" or "dewpoint pressure" (whichever is applicable) rather than "saturation pressure."

Impurities

The impact of impurities on the properties of anthropogenic CO_2 must be considered. While there are processes to remove the impurities, the high cost of purification dictates that some amount of the impurities will remain in the CO_2 transported by pipelines. Common impurities include Ar, N_2, CO, CH_4, H_2, H_2S, O_2, SO_x, NO_x, and H_2O.

Impurities such as O_2, SO_x, NO_x, H_2S, and H_2O can change the susceptibility of pipe steel to corrosion or cracking, damage non-metallic materials and elastomers, and change the toxicity of the system requiring an update to the risk models used during pipeline design. In some cases, experimentation is necessary to determine the allowable levels of these impurities to be able to manage and prevent internal corrosion or cracking of the pipeline. Specific corrosion issues will be discussed in section "Minor Differences".

Small amounts of non-condensable impurities such as Ar, N_2, CO, H_2, and O_2 can modify the maximum pressure of the phase boundary (the cricondenbar). For example, Fig. 1 shows that the maximum pressure of the phase boundary for 100% pure CO_2 (the critical pressure) is 7.39 MPa, and Fig. 2 shows that the maximum pressure of the phase boundary for a 95% pure CO_2 (the cricondenbar) is about 8.6 MPa, an increase of about 1.2 MPa. This increases the pressure required to keep mixtures of CO_2 in the dense phase and to prevent two-phase flow in the pipeline.

Non-condensable impurities also increase the steel toughness required to control running ductile fractures. After the steel toughness required to control running ductile fractures has been specified and the pipe has been procured and installed and the pipeline is operating, the composition and/or the cricondenbar of the source gas must be monitored, and any source gas that requires tougher pipe than has been installed must be locked out to eliminate the possibility of forming a long running ductile fracture. This issue will be discussed at greater length in section "Running Ductile Fractures".

Operating Pressures

Figure 2 shows how the two-phase region of CO_2 cuts through the middle of the normal operating range of hydrocarbon pipelines forcing carbon dioxide pipelines to operate either at high pressure in the dense phase above the two-phase region, or at low pressure in the gas phase below the two-phase region. Operating carbon dioxide pipelines at pressures that are normal for oil and gas pipelines creates two-phase flow regimes that are difficult to manage. Dense phase and gas phase carbon dioxide pipelines each have their own unique advantages and disadvantages, and it is important for the designer to consider the specific requirements of each project before deciding which operating pressure range would be most suitable for the project – dense phase or gas phase.

Carbon dioxide pipelines can be operated safely at high pressure in the dense phase above the two-phase region where the unit cost of transportation is roughly half the unit cost of transportation in the gas phase, making it advantages even though operating dense phase pipelines is more challenging:

1. Wide swings in density caused by relatively small changes in operating temperature require more complex pump/compressor configurations and operations.
2. Careful monitoring of suction pressure and impurities is required to prevent two-phase conditions at the suction of pumps/compressors.
3. Low metal temperatures during both accidental and planned blowdown can lead to more stringent material specifications.
4. Uncertainties associated with controlling running ductile fractures can require specialized pipe testing during design and fabrication.

The challenge mentioned above of handling wide swings in density caused by relatively small changes in operating temperature can be managed but requires attention during the design of booster stations on longer dense phase pipelines. If the suction pressure of a dense phase booster station is 9 MPa and if seasonal fluctuations cause the suction temperature to fluctuate between 15 °C and 30 °C, the density of the CO_2 will vary between 800 kg/m^3 and 500 kg/m^3 (see Fig. 3). This means a 5% change in absolute temperature causes a 40% to 60% change in density. This means the density of dense phase CO_2 is about ten times more sensitive to changes in temperature than an ideal gas, making dense phase CO_2 unlike either a

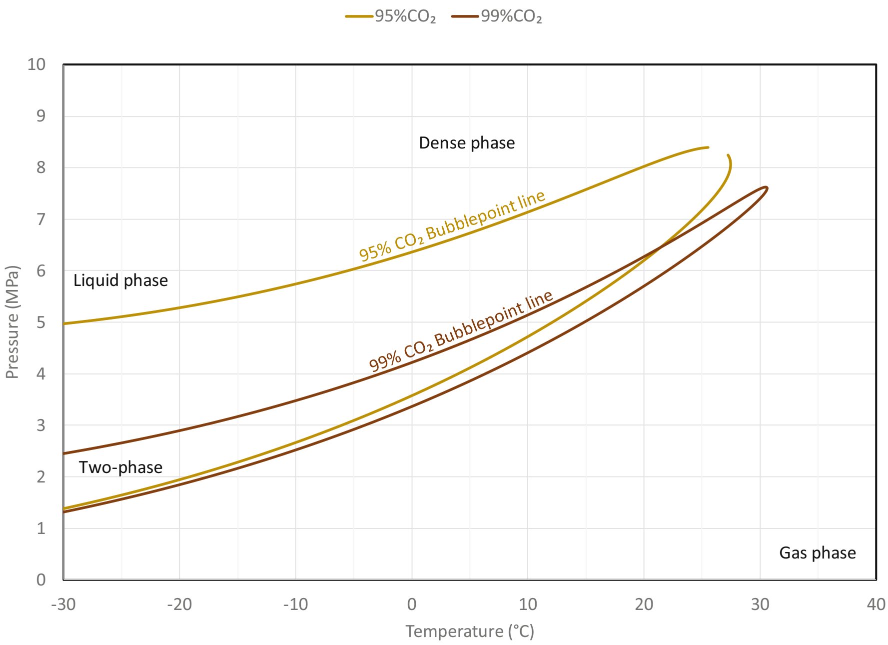

Fig. 3 Effect of impurities such as hydrogen, nitrogen, and methane on the phase boundary of CO_2

gas or a liquid in this regard. Compressor/pump configurations and speed control systems must consider this unusual sensitivity of dense phase CO_2 to temperature fluctuations, whereas gas phase CO_2 pipelines behave in a similar way to natural gas pipelines and do not require innovative compressor configurations and controls.

Repurposed pipelines are better suited for operation at low pressures in the gas phase below the two-phase region for several reasons. Firstly, existing pipelines can seldom handle the high pressures required for economical dense phase CO_2 service. Secondly, operation of gas phase carbon dioxide pipelines is more like the operation of natural gas pipelines because both fluids behave like gases. And finally, gas phase carbon dioxide pipelines do not have the same high toughness requirements required by dense phase carbon dioxide pipelines to control running ductile fractures. One point of concern with this last advantage of gas phase carbon dioxide pipelines is that no full-scale pipe burst tests have been conducted to find out how to calibrate fracture arrest equations for gas phase CO_2.

Fluid Property Predictions

Equations of state (EOS) are used to predict the thermodynamic behavior of pipeline fluids. Because they are not perfectly accurate when predicting thermodynamic properties of CO_2 mixtures, it is important to select a suitably accurate EOS and tune it to experimental data that best represents the CO_2 that the pipeline will carry.

The three most important thermodynamic properties are density, sonic velocity (the speed of sound in the CO_2), and the bubblepoint pressure and temperature near the top of the phase boundary (the cricondenbar).

Zhao et al. compared sonic velocity measurements obtained experimentally with sonic velocity predictions of six equations of state (Zhao et al. 2017):

1. Peng-Robinson (PR).
2. Peng-Robinson with Boston-Mathias alpha function (PR-BM).
3. Soave modified Redlich-Kwong (SRK).
4. Lee-Kesler-Plöcker (LKP).
5. Benedict-Webb-Rubin modified by Starling (BWRS) (Starling and Powers 1970).
6. Span-Wagner (SW) (Span and Wagner 1996).

Zhao et al. concluded that the Span-Wagner EOS is the most accurate of the six equations of state in the single phase near the critical point, and the BWRS EOS is the second most accurate. It should be noted that the Span-Wagner EOS was specifically developed for pure CO_2, while the BWRS EOS is a general-purpose EOS that can predict the properties of mixtures of most gases found in the oil and gas pipeline industry. The Span-Wagner EOS therefore cannot compete in its present form with more generalized equations of state when designing carbon dioxide pipelines carrying mixtures of CO_2 and impurities.

Wagner has been instrumental in developing the general-purpose GERG 2008 EOS which is based on the same fundamental approach as the Span-Wagner EOS. Varzandeh et al. (Varzandeh 2017) compared GERG-2008 and Soave-BWR EOS and concluded that the overall accuracy of Soave-BWR is comparable to GERG-2008 and provides accurate density values near the critical point as well as a better performance than GERG-2008 in bubblepoint pressure and vapor phase composition of binary mixtures. They also found Soave-BWR is much simpler than GERG-2008 and is easier to use for a full range of impurities.

Botros (Botros 2017) compared decompression wave velocities based on empirical pressure measurements of shock tube tests with decompression wave velocities predicted by GERG-2008 and Peng-Robinson. GERG 2008 was found to predict plateau pressure about 1 MPa too high for a binary mixture of CO_2 and CO, and about 0.8 MPa too low for a binary mixture of CO_2 and H_2. Accurate prediction of the plateau pressure is important when calculating the toughness required to control running ductile fractures, and an EOS that sometimes predicts high and sometimes low by wide margins, like GERG 2008 in the Botros tests, is not practical for designing fracture resistant pipe.

Summary

Figure 2 shows how the two-phase region of CO_2 cuts through the middle of the normal operating range of traditional oil and gas pipelines forcing carbon dioxide pipelines to operate at pressures that are either higher or lower than normal for oil and gas pipelines.

New carbon dioxide pipelines can be operated safely and economically at high pressure in the dense phase above the two-phase region, while repurposed pipelines are better suited for operation at low pressure in the gas phase below the two-phase region. Dense phase and gas phase carbon dioxide pipelines each have their own unique advantages and disadvantages, and it is important for the designer to consider the specific requirements of each project before deciding which operating pressure range would be most suitable.

When designing ductile fracture resistant pipe, the designer should understand the advantages, disadvantages, and accuracies of the EOS that will be used to predict density, sonic velocity, and the phase boundary of mixtures of CO_2 with impurities.

Running Ductile Fractures

Running ductile fractures in high-pressure natural gas pipelines have posed serious safety concerns, and the natural gas pipeline industry has invested significant money and resources over the last 60 years to understand the phenomenon and find effective solutions. The first dense phase carbon dioxide pipelines were built in the early 1970s and the possibility that very long running ductile fractures could occur in them was first identified in the late 1970s (King 1979). Nine full-scale pipe burst tests using CO_2 as the test fluid have been conducted to explore pipe wall toughness needed to arrest running ductile fractures in carbon dioxide pipelines, but the number of tests is too small to fully explain the differences between the behavior of running ductile fractures in natural gas pipelines and in carbon dioxide pipelines. Table 5 identifies the full-scale burst tests using CO_2 as the test fluid that have been conducted so far.

Fracture Arrest Equations

In the unlikely event, a dense phase carbon dioxide pipeline bursts, the CO_2 inside the pipe will decompress directly into the two-phase region where the speed of sound, and hence the speed of decompression, is unusually slow. This means that carbon dioxide pipelines depressurize much more slowly than natural gas pipelines and require much greater toughness to control running ductile fractures than natural gas pipelines.

The fracture arrest equations widely used for carbon dioxide pipelines are modified forms of the fracture arrest equations developed for natural gas pipelines in the 1960s and early 1970s for the American Gas Association by Battelle (Maxey et al. 1976). In addition to modifications to constants in the Battelle fracture arrest equations, the thermodynamic properties of CO_2 are used in place of the thermodynamic properties of natural gas.

Table 5 Pipe burst tests using carbon dioxide as the test fluid to determine pipe wall arrest toughness

Test	Date	NPS	Thickness (mm)	Bubblepoint pressure (MPa)	Comment
CO_2 PIPETRANS 1	May 2012	16	6.2, 8.0	6.0	Demonstration of propagation. Propagation observed, arrest at both sides at girth welds
CO_2 PIPETRANS 2	May 2012	16	6.2, 8.0	4.0	Demonstration of arrest. Arrest at both sides in pipe
National Grid 1	May 2012	36	25.4	8.05	Demonstration of propagation and arrest. Propagated longer than predicted
National Grid 2	Oct 2012	36	25.4	7.34	Demonstration of propagation and arrest. Propagated longer than predicted
National Grid 3	Apr 2014	24	19.1	8.9	Demonstration of propagation and arrest. Propagated longer than predicted
SARC02B 1	Feb 2015	24	12.7, 13.7	5.8	Fracture did not propagate beyond initiation pipe
SARC02B 2	Apr 2016	24	12.5, 13.7	7.2	Full propagation to east side. Arrest in last pipe on west side
CO_2 SAFEARREST 1	Sep 2017	24	13.4–13.8	7.8	Propagation on both sides. Arrest in the third pipe on either side
CO_2 SAFEARREST 2					Results not published at time of writing

The nine full-scale pipe burst tests that has been done using CO_2 as the test fluid are not enough to reliably calibrate fracture arrest equations for carbon dioxide pipelines over the full range of pipe size, steel grade, wall thickness, Charpy V-Notch (CVN) toughness, and burial type. Furthermore, full-scale pipe burst tests using natural gas as the test fluid have shown that Battelle's existing fracture arrest equations require arbitrary correction factors for natural gas pipelines made from modern high strength steels (Pistone et al. 1993). In a similar way, problems can be expected when extrapolating the results of CO_2 pipe burst tests. Therefore, reduced-scale laboratory tests (such as shock tube tests) and full-scale pipe burst tests are recommended unless published test results are available for pipe and CO_2 compositions that are the same as the pipe and CO_2 compositions proposed for the project.

Impurities and Toughness Requirements

Figure 3 shows how increasing the amount of common impurities in CO_2 such as hydrogen, nitrogen, and methane cause the two-phase region to balloon up into the dense phase and increase the bubblepoint pressure. The increase in the pressure of the bubblepoint line when the amount of impurities like hydrogen, nitrogen, argon, and methane is increased:

1. Increases the minimum allowable suction pressure needed to avoid two-phase flow in the pipeline.
2. Increases the pressure inside the pipe in the event of a burst, which in turn increases the pipe toughness needed to control running ductile fractures (if one initiates).

Good engineering practice requires the accuracy of pipe wall toughness predictions to be properly understood so that appropriate margins of safety can be calculated. These accuracy calculations focus on the following key areas:

1. Impurities that the pipeline will carry during its operating life that may not be anticipated during design (sometimes referred to as future-proofing the pipeline).
2. Decompression behavior of project specific CO_2 mixtures with particular interest in decompression paths that cross the phase boundary at the cricondenbar and often dictate the required fracture resistance for NPS 20 and larger pipelines.
3. Empirical fracture arrest equations that link pipe wall metallurgy (that is, steel grade, steel toughness, and pipe wall thickness) with crack tip pressure and velocity based on the small number of published full-scale burst tests that use CO_2 mixtures as the test fluid.

Generous margins of safety are recommended to manage the uncertainties inherent in the calculations. Small-scale laboratory tests and full-scale field tests using proposed project fluid compositions and pipe sizes can be used to reduce the margins of safety and hence the wall thickness and toughness of the pipe, which in turn can reduce the cost of the Project. On large projects, tests can reduce the cost of the project by much more than the cost of the tests themselves.

Dense Phase Decompression

Figure 4 shows thermodynamic properties for a 95% CO_2 mixture plotted on a pressure-enthalpy diagram, with decompression paths starting from three different pressures in the dense phase. The thermodynamic properties and decompression paths were developed using the Benedict-Webb-Rubin-Starling eleven parameter equation of state and the Han generalized correlation (Han 1972). Thermodynamic properties in the two-phase region for the fraction of a second it takes for the running fracture to arrest after a burst were based on the homogeneous-heterogeneous

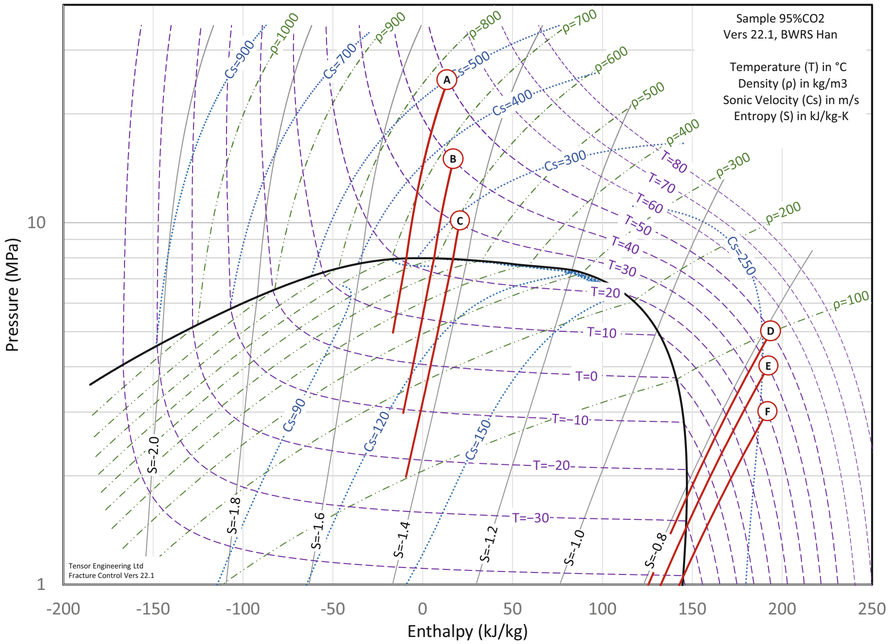

Fig. 4 Thermodynamic properties, phase envelope, and sample decompression paths for 95% CO_2

mixture theory first published in the late 1970s (Han 1972) and now used throughout the pipeline industry for this purpose. The homogeneous-heterogeneous mixture theory is also referred to as the homogeneous-equilibrium theory and the equilibrium assumption.

If a longitudinal ductile fracture initiates, the goal is to arrest it in a short distance (less than eight pipe joints or 100 m). The decompression paths shown in red on Fig. 4 are for the rapid decompression that occurs within the first second of a burst. The calculations assume:

- Frictional effects are negligible because the fracture will arrest in a short distance.
- Heat transfer effects are negligible because the fracture is arrested quickly enough that the total heat flow from the pipe wall to the decompressing fluid is too small to change the temperature of either one.
- The rapid decompression paths shown on Fig. 4 are based on the assumption that friction and heat transfer are both negligible for the fraction of a second it takes to arrest the fracture, so that decompression paths follow lines of constant entropy.

Because heat transfer between the pipe wall and the decompressing CO_2 is negligible in the first fraction of a second, the minimum operating temperature of the fluid in buried sections of the pipeline can be used as the test temperature for CVN tests. Generally, this is taken as the minimum annual temperature at pipeline depth and is usually in the range between −10 °C and + 30 °C depending on where the pipeline is in the world. Above ground, pipelines may need to be tested at temperatures as low as −45 °C in cold regions to guarantee ductile behavior.

Figure 4 shows a sharp discontinuity in sonic velocity at the phase boundary between the dense phase and the two-phase region. For example, decompression Path A starts from operating conditions of 26 MPa and + 50 °C and crosses the phase boundary at about 7.5 MPa and + 20 °C. Sonic velocity in the dense phase just above the phase boundary is about 300 m/s, and in the two-phase region just below the phase boundary is less than 90 m/s. This sharp drop in sonic velocity as the CO_2 decompresses across the phase boundary causes a sharp drop in the velocity of the escaping CO_2, which in turn causes the fluid pressure at the crack tip to be sustained at a high pressure and as a result requires an increase in the pipe toughness to arrest the long ductile fracture.

Decompression from Point C (the lowest of the three initial pressures in the dense phase on Fig. 4) crosses the phase boundary at about the same pressure as decompression from Point A (the highest of the three initial pressures). This means that decompression from Point C will have the same sustained pressure plateau at the crack tip as decompression from Point A and can be expected to require the same toughness to arrest a longitudinal ductile fracture.

Sometimes decompression from the lowest pressure can require the highest toughness, which is counterintuitive and introduces an unusual design opportunity in which the minimum operating pressure dictates yield strength, wall thickness, and toughness based on running ductile fracture control equations, and the maximum operating pressure can then be back-calculated from these parameters using conventional methodology such as limit states design or the Barlow formula combined with design factors as allowed by governing codes or regulations.

Two Curve Methods for CO_2

The control of longitudinal ductile fractures in carbon dioxide pipelines using pipe wall metallurgy is usually based on modified versions of the Battelle Two Curve Method (TCM), which was originally developed for natural gas pipelines. As its name implies, the Battelle TCM uses two curves:

1. The Pressure Wave Velocity Curve (solid lines on Fig. 5) predicts the speed at which gas pressure inside the pipeline decays after a burst and is based on an equation of state and a high-speed transient flow equation.
2. The Fracture Velocity Curve (dashed lines on Fig. 5) predicts the speed of the fracture as it is driven along the pipe by the pressure of the fluid at the crack tip and is based on the Battelle arrest pressure equation and Battelle fracture velocity equation.

The toughness needed to control long running fractures is found from the Fracture Velocity Curve when it just touches the Pressure Wave Velocity Curve. Modified

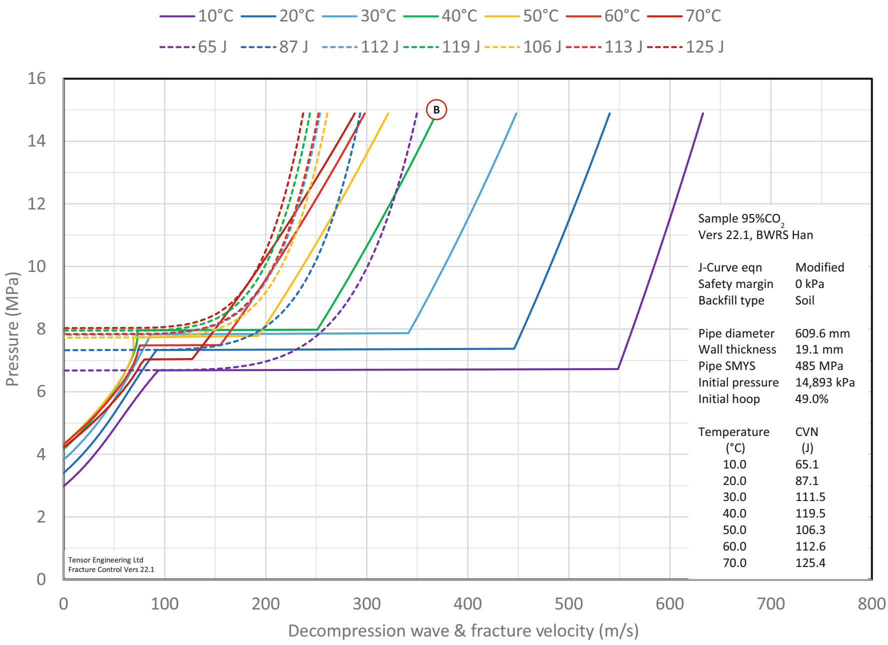

Fig. 5 Pressure wave velocity (solid line) and fracture velocity (dashed line) curves for NPS 24 X70 pipe with 19.1 mm wall thickness at 14.65 MPa

TCMs for carbon dioxide pipelines use the same form of equations as the Battelle TCM but the constants must be recalibrated to make predictions fit either:

1. Results of full-scale pipe burst tests using project-specific pipe and project-specific CO_2 compositions.
2. Published results of pipe burst tests (see Table 5) if project pipe parameters and CO_2 compositions under consideration do not lie outside the range of parameters used in the published pipe burst tests.

Attempts have been made to develop finite element methods using fundamental fracture behavior models, computational fluid dynamics, DWTT values instead of CVN values, and coupled equations instead of the uncoupled TCM developed by Battelle. Although these potentially more precise models have produced promising results, they still require to be validated against empirical test results and have not yet been developed far enough to produce better predictions than modified versions of the Battelle TCM.

Figure 5 shows the two curves of a modified TCM for NPS 24 X70 pipe with wall thickness of 19.1 mm operating at a pressure of 14.9 MPa for a range of operating temperatures between 0 °C and 60 °C. The Pressure Wave Velocity Curves are represented by solid lines and the Fracture Velocity Curves are represented by dashed lines. Point B on Fig. 5 corresponds to Point B on Fig. 3.

Gas Phase Decompression

The set of published full-scale pipe burst tests listed in Table 5 do not include any tests on gas phase carbon dioxide pipelines. There is, therefore, no empirical burst test data that can be used to calibrate TCMs for gas phase carbon dioxide pipelines. One method for gas phase carbon dioxide pipelines is to use a TCM calibrated for dense phase carbon dioxide pipelines. Although this method is likely to produce conservative results, it is not recommended because it does not comply with natural gas pipeline industry requirements such those included in API Specification 5L, Annex G, Note 2 that requires designers to “take all reasonable steps to ensure operating parameters, including gas composition and pressure, of any gas pipeline to which the requirements of this Annex apply are comparable or consistent with the test condition on which the . . . method was established.”

Figure 6 illustrates results for a gas phase pipeline using a modified TCM with large margins of safety to provide for unknowns associated with fracture resistance of gas phase carbon dioxide pipelines. Point D on Fig. 5 corresponds to Point D on Fig. 3. Reductions in the margins of safety could be achieved if gas phase tests were undertaken. For large projects, the resulting reductions in wall thickness and toughness will reduce project costs by more than the cost of the tests themselves.

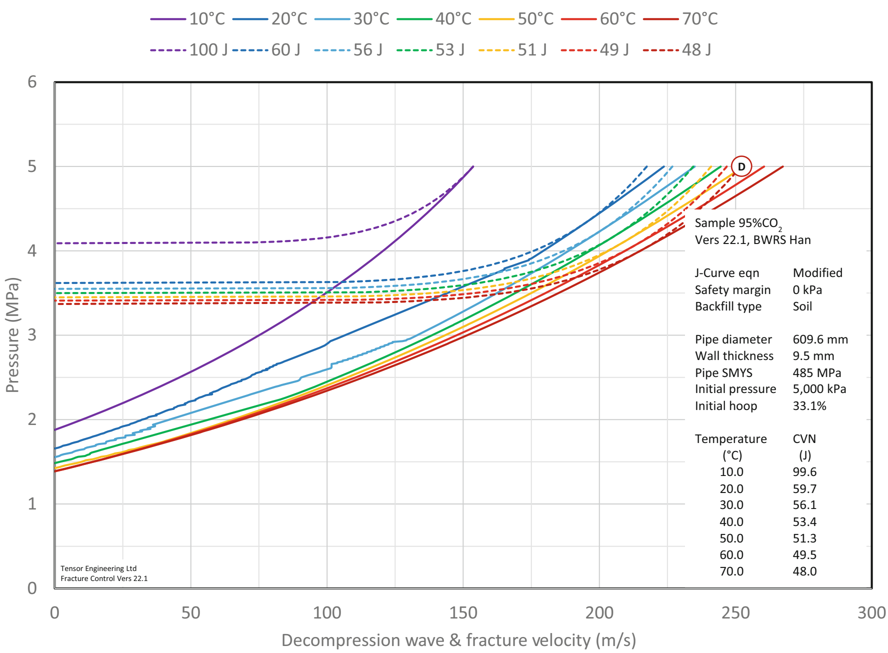

Fig. 6 Wave velocity (solid lines) and fracture velocity (dashed lines) curves for NPS 24 X70 pipe with 9.5 mm wall thickness at 5 MPa

Pipe Materials

After selecting yield strength, wall thickness and toughness to prevent initiation of bursts and to control running ductile fractures as already discussed, there are no other major differences in the design and specification of pipe between carbon dioxide pipelines and natural gas pipelines. But the recent failure of a carbon dioxide pipeline (US Department of Transportation, Pipeline and Hazardous Materials Safety Administration 2020) has raised public concern about the safety of carbon dioxide pipelines that are being taken seriously by the pipeline industry.

The primary cause of the failure was subsidence of the right of way after heavy rain and a secondary cause was localization of strain in a girth weld due to softening of the weld area along with actual lower strength weld metal compared with the actual tensile strength of the pipe material which caused the weld to be undermatched. Specific lessons learned from this event are the need for early identification of geotechnical hazards, for implementing overmatched girth welds, and for accurately delineating the impact zone along the full length of the route.

Although the cause of this failure was unrelated to the contents of the pipeline or to any difference between carbon dioxide and natural gas pipelines, it is interesting to observe that problems with undermatched girth welds have recently been experienced on oil and gas pipelines. This highlights the importance of staying abreast of new problems encountered on natural gas pipelines and to transfer the solutions to the design, construction, and operation of carbon dioxide pipelines.

Problems associated with softening of the HAZ and undermatched weld metal compared to the actual strength of modern high strength, high toughness pipe steels are relatively new for the pipeline industry. The problem has been solved by modifying girth welding procedures, enhancing quality control practices, managing actual maximum pipe tensile strength, adding specific procedure qualification tests of the weld area, and enhancing weld procedure controls. The overmatching of girth welds is dealt with in greater detail in section "Field and Fabrication Shop Welding" on welding.

Pipe for carbon dioxide pipelines typically conforms to API Specification 5L and CSA Z245.1 Steel Pipe. Pipe may be procured seamless or with a seam that is either longitudinally or helical welded. Longitudinally welded pipe may be joined either with electric welded (EW) process or submerged arc welded (SAW) process while helical pipe is welded using submerged arc welded process. The pipe body and weld (where applicable) must be tested for adequate wall thickness, yield strength, tensile strength, toughness, heat treatment state, and in some cases hardness and chemistry. The parameters of greatest interest for carbon dioxide pipelines that have not been covered in previous sections are:

1. Heat treatment state must be known before girth welds can be designed to prevent softening of the weld area and hydrogen cracking.
2. Chemistry and actual tensile strength must be known for pipe weldability purposes and to correlate with steel microstructure to understand its resistance to sulfide stress cracking and corrosion.

In areas with dense populations or geotechnical hazards, consideration should be given to ensure the CVN toughness of the long weld and the HAZ is equal to or better than the CVN toughness of the pipe body. It should be noted that only a handful of manufacturers can make EW welds with high CVN values at the fusion line due to the nature of inclusions formed during electric resistance welding.

A CVN test temperature equal to the lowest anticipated operating temperature should be specified and to ensure that running fractures will be ductile rather than brittle, the pipe body should be tested using a drop weight tear test (DWTT) at the lowest anticipated operating temperature. A shear area of 85% or higher (the Battelle criterion) should be specified.

Repurposing Existing Pipelines

DNV-RP-104 recommends that pipelines repurposed to carry CO_2 should satisfy all the requirements for new carbon dioxide pipelines unless an engineering assessment is undertaken in accordance with code requirements to show that the deviations from code are acceptable (DNV GL 2019). The approach is "proof of concept" via operating pressure assessment followed by a hydraulic assessment prior to undertaking detailed engineering on the condition of the pipeline and any potential upgrades required for conversion.

Operating Pressure

Most pipelines available to be repurposed for carbon dioxide service are not made from high toughness steels and have ANSI 600 flanges that restrict the maximum operating pressure to 9.93 MPa or less. The low toughness and narrow operating range between 9.93 MPa and the bubblepoint of CO_2 mixtures, makes pipelines with ANSI 600 flanges better suited for gas phase CO_2 service at pressures below approximately 4.2 MPa than for dense phase CO_2 service.

A few existing pipelines have ANSI 900 flanges and are designed to operate at pressures up to 14.89 MPa, making them suitable for conversion to dense phase CO_2 service with the addition of crack arrestors if they do not have the toughness required to control running ductile fractures. The unit cost of service of new pipelines that operate at low pressure in the gas phase beneath the two-phase region is approximately twice the unit cost of service of new dense phase carbon dioxide pipelines, but if existing pipelines can be acquired for a low enough price, they can be repurposed for gas phase CO_2 service and compete economically with new dense phase pipelines. In addition, they have advantages over dense phase pipelines for operation in highly populated and congested areas:

- Pipe bursts do not have the same force as dense phase pipe bursts and the release volume is less.
- They operate like natural gas pipelines.

Composite Materials

Another potential method of repurposing existing infrastructure is to utilize the retired pipeline as a conduit for pulling composite pipe. This can be a feasible cost-competitive alternative in highly congested areas where construction of a new dense phase carbon dioxide pipeline is impractical. Table 6 lists several products that have been designed for higher pressures and temperatures, which would enable the transportation of dense phase CO_2 These products have the potential to be used for dense phase carbon dioxide with further due diligence on the potential failure mechanisms associated with both the metallic and non-metallic components.

Low Temperatures due to Leaks and Rapid Depressurization

When dense phase carbon dioxide pipelines depressurize during pipe bursts, or rapid blowdown, they get cold towards the end of the depressurization and form a drift of dry ice in the bottom of the pipe. The temperature of the dry ice is close to −80 °C and the cooling effect can draw down the average through-wall temperature well below its transition temperature. A similar phenomenon occurs in the vicinity of leaks. The temperature of the escaping CO_2 falls to around −80 °C and can draw down the temperature of the pipe wall below its transition temperature. In both cases, concern has been expressed that this hypothetical mechanism could initiate a running ductile fracture, and many carbon dioxide pipelines have been designed with minimum metal temperatures of −45 °C and lower.

Table 6 Commercially available spoolable technologies potentially capable of transporting dense phase CO_2

Manufacturer	Product name	Reinforcement type	NPS	Pressure range (psig)	Pressure range (MPa)
FlexSteel[a]	FlexSteel Line Pipe	Helical steel strips	2-inch to 8-inch	2250	15.51
			2-inch to 6-inch	3000	20.68
Smart Pipe[a]	SmartPipe Line Pipe	Fiberglass tape	4-inch to 16-inch	>2000	> 13.79
Baker Hughes	Composite Pipe	Fiberglass tape	2-inch to 6-inch	750–2,250	5.17–15.51
SoluForce	Heavy	Glass Fiber tape	6-inch	5627	38.80
TechnipFMC	CoFlexip	Steel hose	As designed	As designed	

[a]Products with monitoring technologies/methods are highlighted in light grey

Warm Prestressing

This cooling mechanism has not been found to cause failures in natural gas pipelines and it is not normally considered during design, even though natural gas can cool below −160 °C when it decompresses. One reason is that the walls of natural gas pipelines do not get as cold as carbon dioxide pipelines because natural gas is less dense and has less cooling effect on the pipe steel. Another reason is the beneficial effect of warm prestressing on defects in the pipe wall. Warm prestressing occurs when a load is applied to a structure containing a defect while its temperature is above its transition temperature and then the temperature of the structure is lowered below its transition temperature. Warm prestressing rounds out the sharp edges of the defect and causes the structure to fail at a higher load at temperatures below its transition temperature than it otherwise would.

Experiments using single edge notch bend tests have been conducted to demonstrate the beneficial effect of warm prestressing on linepipe steel (Cosham et al. 2013). Single edge notch bend specimens were subject to "load-cool-fail" and "load-unload-cool-fail" cycles. The load at failure of the warm prestressed specimens was higher than those that had not been warm prestressed, and in most cases load at failure was higher than the pre-load which proved that cooling the pipeline below its transition temperature will not result in failure providing the load at low temperature is not greater than the warm prestressing load.

Therefore, longitudinal fractures will not initiate when the temperature of the pipe steel falls below its transition temperature because the hoop stress acting on longitudinal defects at low temperatures will always be less than the hoop stress that the defect has already experienced at normal operating temperatures.

Circumferential Defects

The same cannot be said for circumferential defects in the girth welds exposed to axial tensile forces. Thermal contraction can apply larger axial tensile stresses across girth welds than most of the welds have previously experienced while they were warm and ductile under normal operating conditions. This sequence of events, where the low temperature load is greater than the warm prestressing load, can cause circumferential cracks in the girth welds to grow and possibly cause problems at the welds when the pipeline is repressurized to return it to service.

A workaround that is sometimes suggested is to run an ILI crack tool after a pipe burst or rapid blowdown to detect circumferential cracks and repair any that are identified. But detection of circumferential cracks in welds using ILI tools can be difficult and unreliable. Ultrasonic (UT) ILI crack tools with a reasonable probability of detection (POD) for finding circumferential cracks require the tool to be run in a slug of liquid couplant (diesel or equivalent) and they currently have a wall thickness limitation of 25 mm which can be a problem for larger diameter dense phase carbon dioxide pipelines. Electromagnetic Acoustic Transducer (EMAT) ILI tools can find circumferential cracks without a liquid couplant, but they have more stringent wall thickness limitations than UT tools and a lower POD due to signal attenuation, dispersion, and interference making them less reliable for finding circumferential cracks in girth welds.

A better solution is to determine the lowest metal temperature that can occur in the pipeline after a pipe burst or rapid blowdown and the resulting axial stress due to thermal contraction. Welding consumables with adequate strength and toughness at the lowest metal temperature can then be specified and the maximum allowable circumferential defect size can be calculated for use in the weld qualification procedure and welding QA/QC processes.

Blowdown Stacks and Repressurization

Blowdown stacks get cold during depressurization and pipelines downstream of the throttling valve get cold when pipelines are repressurized following a repair, or during initial fill from high-pressure sources, such as the upstream segment. Natural gas pipelines share this problem and procedures and metallurgy similar to those used for natural gas pipelines can be used for carbon dioxide pipelines except that the thermodynamic properties of CO_2 are used in place of natural gas.

Field and Fabrication Shop Welding

Typical welding processes include Shielded Metal Arc Welding (SMAW), Flux Cored Arc Welding (FCAW), Gas Metal Arc Welding (GMAW), Metal Cored Arc Welding (MCAW), and Sub-Arc Welding (SAW). Each welding process has a different setup and requires a variety of welding skills, welding equipment, and welding consumables.

Most of the challenges associated with welding carbon dioxide pipelines are the same as welding natural gas and crude oil pipelines. Welding challenges include attaining sufficient microstructure to prevent hydrogen cracking, overmatching the weld to the base material, and achieving sufficient notch toughness.

The codes and standards governing the welding of girth welds include CSA Z662, ASME IX, and API 1104. These codes and standards are meant to be minimum requirements and a welding engineer experienced in pipeline girth welding should be consulted. When hydrogen sulfide (H_2S) is expected in the product stream, NACE MR0175 should be consulted, and when amine is expected in the product stream, API RP 945 should be consulted.

Fabrication Welds

While CSA Z662 and API 1104 may be used, ASME IX is generally used for new construction fabrication welds. Fabrication shop welding can use positioning equipment that allows welds to be “rolled” and enables higher rates of weld deposition. In these shops, low hydrogen welding consumables are usually preferred and readily available, helping to reduce hydrogen cracking and making it easier to overmatch the welds, especially when the root pass is made with low hydrogen consumables.

Pipeline Girth Welds

Most welding procedures for buried natural gas pipelines incorporate either no toughness testing or toughness testing down to −29 °C at the lowest. This enables most natural gas operators to specify cellulosic weld consumables to gain productivity and ease of welder training as these types of consumables are easier to use compared to low hydrogen.

However, rapid depressurization of dense phase CO_2 may expose girth welds to dry ice and temperatures approaching −80 °C. When this happens, the temperature differential between the construction lock-in temperature and − 80 °C may cause pipe steel to experience significant thermal contraction stress, especially if the pipeline was installed and locked in during the summer.

Figure 7 shows the effect of low metal temperature on the longitudinal stress of a buried pipeline made from Grade 448 (X65) steel. In this case, weld consumables that have been tested to temperatures lower than −30 °C could be considered. This would require the use of low hydrogen welding consumables and in many cases, the use of welding consumables either supplementary batch/lot tested or qualified by the manufacturer to lower temperatures such as −50 °C or − 60 °C. In addition to the consideration of blowdown temperatures, the girth weld should exhibit toughness properties like that specified for initiation in base material. This may require the specification of either higher absorbed energy values or lower testing qualification temperatures of the weld and HAZ.

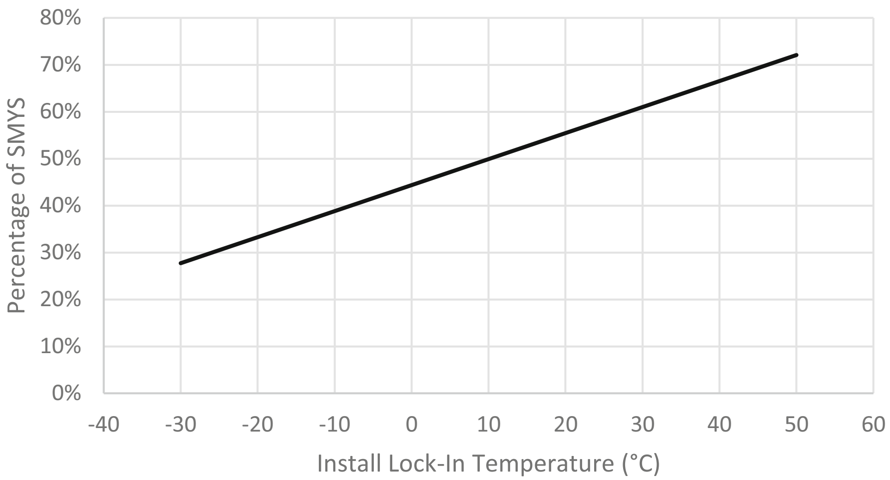

Fig. 7 Potential longitudinal stress due to thermal contraction during blowdown for Grade 448 pipe

Methods that would help enable higher toughness values at lower temperatures include:

- Use lower heat input passes and require stringer techniques.
- Use weld consumables qualified to lower temperatures, which usually are low hydrogen weld consumables.
- Use welding processes that enable the deposition of welds with greater toughness.
- Reduce or eliminate the use of cellulosic weld consumable in the root and hot pass.
- Increase the amount of toughness qualified weld consumable in the weld joint, and increase the thickness of the girth weld (if possible).
- Use several multiple passes during welding.
- Restrict the maximum interpass temperature in the weld procedure.

Another potential option would be to modify the acceptance criteria of weld defects by performing an Engineering Critical Analysis, as defined in CSA Z662 Annex K (CSA Group 2019).

Weld Overmatching

To prevent failure in the weld area, weld overmatching is desirable to distribute longitudinal strain throughout the pipe body and prevent it from concentrating in the weld area. This means that the actual weld metal strength and actual HAZ strength should be higher than the base material longitudinal strength. Undermatching can occur either in the weld metal, the HAZ, or both.

Methods to ensure overmatching is achieved include:

- Selecting a welding procedure that limits HAZ softening, such as lowering maximum heat input, reducing size of HAZ (use of low hydrogen consumables), specifying joint angle to prevent failure in the HAZ.
- Selecting a weld consumable that generally has higher chance of overmatching the actual tensile strength of the base material.
- Specifying a geometric overmatch as a method to manage weld area softening.

Welding Consumable Selection

Typical welding consumables used for pipelines are shown in Table 7. It should be noted that consumable selection may not always overmatch actual base material tensile strength, nor achieve adequate toughness at lower temperatures. Consumables may need to be specially tested by suppliers or lot/batch tested if toughness at lower temperatures is found to be necessary.

If existing welding procedures are used, a thorough evaluation must be made to ensure their adequacy for carbon dioxide pipelines which may include ensuring the

Table 7 Table of typical welding consumables[a]

Welding process	Welding consumable: Root pass	Welding consumable: Hot pass, fill & cap	SMYS of electrode (MPa)	SMTS of electrode (MPa)	Suggested base material grade[b]	Toughness of fill/cap[c]
SMAW	E6010	E7010-P1	330 / 400	430 / 490	Grade 386 and Lower	27 J @ –30 °C
SMAW	E6010	E8010-P1	330 / 400	430 / 490	Grade 386 and Lower	27 J @ –30 °C
SMAW	E6010	E7018–1	330 / 400	430 / 490	Grade 386 and Lower	27 J @ –45 °C
SMAW	E6010	E8018-C3	330 / 470	430 / 550	Grade 359 to 414[d]	27 J @ –40 °C
SMAW	E8010-P1	E8018-C3	460 / 470	550 / 550	Grade 386 to 448[d]	27 J @ –40 °C
SMAW	E8010-P1	E9045-P2	460 / 530	550 / 620	Grade 386 to 483	27 J @ –30 °C
SMAW	E8010-P1	E10045-P2	400 / 600	430 / 690	Grade 414 to 550	27 J @ -30 °C
GMAW-S/ FCAW	ER70S-6[e]	E81T1-Ni1C-J	400 / 470	490 / 550	Grade 386 to 448[d]	27 J @ –40 °C
SMAW/ MCAW	E7010-P1	E90C-K3	400 / 540	430 / 620	Grade 386 to 483	27 J @ –50 °C
SMAW/ MCAW	E8010-P1	E100C-K3	460 / 610	550 / 690	Grade 414 to 550	27 J @ -30 °C

[a] Listed options are a select set of welding processes and welding consumables and several other variations exist which in addition to those listed would have to be evaluated for a given application
[b] Weld overmatching may still not be achieved as actual tensile strength of the base material grade may be greater than the SMTS of the electrode. Many other factors exist that may govern the need for overmatch and the level of overmatch solely based on weld metal. This table does not address HAZ softening issues that can also contribute to potential cross weld undermatching
[c] Options may exist with certain manufacturers that perform toughness testing at lower temperatures to certify the consumables
[d] Options may exist for up to Grade 483 for thicker pipe materials
[e] Low hydrogen root pass generally utilizing short circuit or modified short circuit, such as Lincoln Surface Tension Transfer® (STT) or Miller Regulated Metal Deposition (RMD™)

welds overmatch the actual pipe tensile strength, especially in areas where potential high longitudinal loads could persist, and the required toughness at lower temperatures can consistently be achieved. This applies independently of the code (ASME IX, CSA Z662, API 1104, or others) used for welding.

Fatigue

Fatigue due to repeated load cycling affects the longevity of carbon dioxide pipelines in different ways from conventional oil and gas pipelines and more research is needed to quantify how the differences impact design.

Amplitude and Frequency

Pump start-up and shut-down, valve operation, flow rate changes, and pressure surges typically cause more frequent and larger pressure fluctuations in liquid pipelines than in gas pipelines and can lead to greater fatigue damage over time. From Fig. 3 showing the thermodynamic properties of a 95% pure CO_2 mixture, isentropic compression from 10 MPa to 25 MPa along Line A, increases the density of CO_2 from about 690 to 780 kg/m^3 giving dense phase CO_2 an isentropic bulk modulus of about 130 MPa.

For comparison, the isentropic bulk modulus of oil and water is between 1.5 and 2.2 GPa, more than 10 times higher than dense phase CO_2. And the isentropic bulk modulus of air and natural gas at 10 MPa is approximately 14 MPa, about 10 times lower than dense phase CO_2. Because the bulk modulus of dense phase CO_2 is between that of oil and gas, fatigue in dense phase CO_2 pipelines can be expected to be less than in oil pipelines, but it should still be considered during design of the system because it is greater than in gas pipelines.

In summary, fatigue in gas phase CO_2 pipelines is expected to be the same as fatigue in natural gas pipelines, whereas the net effect on fatigue and longevity due to the differences between dense phase carbon dioxide pipelines and conventional oil pipelines needs to be explored more fully to understand design guidelines to assure the design life expectancy of dense phase pipelines.

Fire and Asphyxiation Hazards

The relative overall safety of carbon dioxide pipelines is similar to natural gas pipelines, but the public perceives the risk of asphyxiation with alarm, perhaps because the risk is new and unfamiliar rather than the actual level of risk involved. In either case, whether the pipeline carries CO_2 or hydrocarbons, the goal of safety in design is to reduce both actual and perceived risks to the public to as low as reasonably practicable (ALARP).

Case Study

A carbon dioxide pipeline burst in Yazoo County, Mississippi on February 27, 2020 (US Department of Transportation, Pipeline and Hazardous Materials Safety Administration 2020). Approximately 300 people were evacuated and 45 people were treated at area hospitals. The Clarion-Ledger reported that the CO_2 released into the air made people “act like zombies” and that a first responder rescued three people before he too was overcome by the gas. This event has attracted public attention in the USA and prompted a critical review of the safety of carbon dioxide pipelines and the adequacy of Pipeline Hazardous Materials Safety Administration (PHMSA) regulations.

Fire

The lower and upper explosive limits (LEL and UEL) of natural gas when mixed with air are 4% and 16% respectively, and when natural gas pipelines leak or rupture, the escaping gas can catch fire and cause dangerous jet flames, fireballs, and vapor cloud explosions (CSA Group 2018).

Unlike natural gas, CO_2 is non-flammable and does not form an explosive mixture when mixed with air so that carbon dioxide pipelines do not need to manage the consequences of jet flames, fireballs, and vapor cloud explosions. This reduces the risks of carbon dioxide pipelines and makes them inherently safer than conventional oil and gas pipelines in this respect.

Although it is an obvious advantage of carbon dioxide pipelines, it is important to specifically include this advantage in risk assessments that compare the risks of carbon dioxide pipelines with the risks of conventional gas pipelines with which the public has more real-world experience.

Asphyxiation

CO_2 and natural gas are both asphyxiants, but CO_2 is about 50% heavier than air at atmospheric pressure and can accumulate in low-lying areas which increases the risk of asphyxiation, whereas natural gas is about 50% lighter than air and dissipates upwards without creating a credible risk of asphyxiation. The problem is that CO_2 is heavier than air, and under low wind conditions, it can settle in low-lying areas, displace the air, and asphyxiate animals and people. In this regard, carbon dioxide pipelines are less safe than natural gas pipelines.

Strategies to minimize risk of asphyxiation associated with carbon dioxide pipelines include minimization of release volumes by installing check valves and fast-acting automatic isolation valves at shorter intervals than the maximum allowed by code, improving leak detection systems, delineating pipeline impact corridors based on terrain and detailed dispersion modelling, CO_2 monitoring in highly populated and low-lying areas, implementing community awareness programs and emergency response plans, as well as improving vent stack design to increase mixing of vented CO_2 with atmospheric air during pipeline blowdown.

Release Likelihood

Release frequency can be reduced by implementing a well-planned integrity maintenance program and by close monitoring of fluid quality to prevent internal corrosion.

Release Volume

It is common practice during the design of pipelines to overbuild capacity with the goal of producing a conservative design that will easily meet throughput targets. This policy of oversizing the diameter of pipelines increases risk because it increases the stored volume of CO_2 which then increases the size of the plume and the dispersion radius.

With dense phase pipelines, it is usually possible to reduce diameter and increase design pressure to meet throughput targets without increasing the toughness required to control running ductile fractures. This approach has the advantage of making the release volume smaller, the dispersion radius shorter, and the pipeline safer. Release volume and the size of the plume can be further reduced by:

- Optimizing the distance between sectionalizing valves.
- Installing fast-acting automatic actuators at all sectionalizing valves.
- Drawing down pipeline contents before blowdown.

Pipeline Planning Corridor

The lateral extent of the pipeline planning corridor should be determined from detailed dispersion studies that take account of the details of topography, wind speed, and wind direction anticipated throughout the year in each geographical unit along the route. The topographical data must include valleys that can channel the CO_2 long distances from the pipeline route, and low-lying areas where CO_2 can accumulate. The resulting planning corridor should not have a fixed assessment width but should vary according to the geographical features and anticipated weather conditions along the route. A weakness in using a fixed assessment width is that CO_2 can accumulate well outside the fixed assessment corridor in low-lying areas downslope from the pipeline.

Improved Leak Detection

Adverse impacts to the public can be reduced by implementing advanced leak detection systems, such as fiber optic cables installed in the ditch alongside the pipe to detect changes in temperature, and CO_2 sensors installed in communities and low-lying terrain within the pipeline planning corridor along the full length of the route.

Vent Stack Design

Before blowdown of a section begins, preliminary procedures like those used for natural gas pipeline blowdown can be used to reduce the release volume. These include drawing down the pressure in the pipeline to the lowest level possible using the mainline pumps/compressors, and then after the section to be blown down has been isolated at the upstream and downstream block valves, a specially designed mobile pump/compressor units can be used to move the CO_2 around the downstream block valve into the downstream section. Another possible procedure is to displace the CO_2 into the next downstream segment of the pipeline using gel plugs and nitrogen and then vent the nitrogen directly to atmosphere.

These preliminary procedures to reduce the release volume are costly and take time to implement and would not be needed if vent stacks and procedures are designed to safely vent the CO_2 directly to the atmosphere. A goal when designing blowdown stacks for carbon dioxide pipelines is to ensure sufficient mixing of the CO_2 as it is released into the atmosphere and to ensure the low temperatures associated with blowdown are managed such that contraction stresses do not cause undue brittle fracture in the pipe body and girth welds.

Blowing down slowly enough that pipe wall temperatures are kept above the transition temperature of the pipe steel takes too long to be practical and is unnecessary because of the beneficial effect of warm prestressing that was discussed in section "Low Temperatures Due to Leaks and Rapid Depressurization". By the time the temperature of the CO_2 falls below the transition temperature of the pipe steel, the hoop stress due to internal pressure will be so low that initiation of brittle fractures will not be possible. For example, during decompression from 15 MPa and 40 °C (Point B on Fig. 3), by the time the temperature falls to −20 °C, the pressure will have fallen to 2.2 MPa, approximately 15% of the initial pressure and less than 10% SMYS. The only concern is with axial thermal contraction stresses acting across the girth welds when the internal pressure falls low enough to form dry ice inside the pipeline. As already discussed in section "Fatigue", girth welds can be designed to manage anticipated low metal temperatures without aggravating circumferential defects in the girth welds.

Engineered vent stack designs that ensure sufficient mixing of CO_2 with atmospheric air to make blowdown safe are not available in the open literature, and therefore there is no readily available industry guidance. Detailed thermodynamic analysis, supplemented by reduced scale or full-scale testing is recommended to explore possible ways of building blowdown stacks that can maximize the degree of mixing of CO_2 with atmospheric air, and thereby reduce the risk of asphyxiation during blowdown.

Corrosion

Overview

A mixture of CO_2 and water can form carbonic acid (H_2CO_3). The rule for preventing corrosion in pipelines carrying naturally occurring CO_2 has been to prevent free water from precipitating in the pipeline under any operating condition in combination with stringent safeguards to prevent CO_2 with water content exceeding specifications from entering the system. But anthropogenic CO_2, which typically contains H_2S, sulfur oxides (SO_x), and nitrous oxides (NO_x), can cause corrosion even with very low water content levels, far below the level at which free water can form. Based on the threat of corrosion from a combination of CO_2 and its impurities, strict requirements are required on dewatering and drying pipelines after pressure testing (CSA Group 2019; American Society of Mechanical Engineers 2022). This is particularly important for valve components and fittings where trace water can be easily trapped.

Waterless Corrosion

Autoclave experiments have identified and quantified mechanisms by which CO_2 containing combinations of H_2S, SO_x, NO_x, and O_2 can cause corrosion of steel even if the CO_2 contains no water (Morland et al. 2019). Chemical reactions that drive the mechanism are:

$$3NO_2 + H_2S \rightarrow H_2O + SO_2 + 3NO \tag{1}$$

$$H_2O + SO_2 + NO_2 \rightarrow H_2SO_4 + NO \tag{2}$$

$$4NO + 2O_2 \rightarrow 4NO_2 \tag{3}$$

An experiment with different combinations of H_2O, H_2S, NO_x, SO_x, and O_2 revealed that the presence of 35 ppmv each of SO_2, O_2, H_2S, and NO_2 results in formation of a separate aqueous phase containing sulfuric and nitric acid, which is highly corrosive. If either H_2S or particularly NO_2 are removed, these reactions do not occur (Morland et al. 2021). In a related experiment, CO_2 containing approximately 20 ppmv each of H_2S, NO_2, SO_2, and O_2 resulted in chemical reactions (oxidation of H_2S to SO_2), but a separate aqueous phase did not form and there was no corrosion (Svenningsen and Morland 2021). It can therefore be concluded that 20 ppmv each of H_2S, NO_x, SO_x, and O_2 in CO_2 will not cause corrosion.

Experimentation is ongoing to explore corrosive mechanisms that are possible in pipelines carrying anthropogenic CO_2, and it may be some time before the full picture emerges and firm limits on the allowable amounts of H_2O, H_2S, SO_x, NO_x, and O_2 are established. In the meantime, keeping H_2O, H_2S, NO_x, SO_x, and O_2 at low levels is recommended, and pipeline designers should seek expert advice on safe levels according to the latest research available at the time the design work is being undertaken.

H_2S and Sulfide Stress Cracking

The recommended maximum concentration of H_2S in anthropogenic carbon dioxide pipelines recommended in DNV-RP-F104 (DNV GL 2019) is 100 ppmv. A joint industry project code-named "CO_2 Safe and Sour" began in March 2022 with an anticipated time horizon of two years. DNV manages the project in association with Equinor, Shell, Gassco, Total Energies, IFE, Wood, Gassnova. ExxonMobil, Woodside, ArcelorMittal, Vallourec, Subsea7, Tenaris, and Corinth.

The project aims to investigate the integrity of carbon dioxide pipelines used for CCUS applications with a focus on the challenges posed by H_2S in the CO_2 stream. The findings will help determine the impact of H_2S levels on the risk of sulfide stress cracking (SSC) and corrosion damages in carbon steel pipelines. Ultimately, the project aims to update DNV-RP-F104 and provide general recommendations for industry-wide use.

Until results of ongoing experiments are available and firm limits on allowable concentrations of H_2S, NO_x, SO_x, O_2, and H_2O are established for anthropogenic CO_2, strict control of these components at low levels is recommended.

Minor Differences

Successful operation of carbon dioxide pipelines requires careful attention to the minor differences between carbon dioxide pipelines and natural gas pipelines.

Inhibitors

Oil-based corrosion inhibitors are preferred in natural gas pipelines for long-term protection because they have significant film formation and hydrophobic properties. But they are soluble in CO_2 making them ineffective for carbon dioxide pipelines. Keeping the water content at a low level (for example, less than 2 lb./Mscf or 40 ppm) is the primary means of corrosion inhibition in carbon dioxide pipelines, but in unusual circumstances where corrosion inhibition is needed, non-oil-based inhibitors (silicate, nitrites, molybdates, phosphates, zinc salt, etc.) should be considered.

Non-metallics

Critical non-metallic components in natural gas pipeline systems include O-rings, seals, valve seats, gaskets, maintenance pigs, etc. CO_2 can penetrate most polymerics, particularly at high pressures. The severity of penetration in non-metallic components causes an array of degradation mechanisms, particularly upon decompression. Non-metallic materials must be qualified to ensure:

1. Ability to resist swelling and rapid gas decompression (e.g., O-rings, seals, valve seats, pigs, ILI tools).
2. Chemical compatibility with the CO_2 stream without causing decomposition.
3. Hardening or significant negative impact on key material properties.
4. Resistance to the design temperature range.

For inline inspection tools, the rate of degradation of electrical wiring generally limits travel time to less than 48 h, and the wiring needs to be refurbished at the end of each run.

Lubrication

Unlike natural gas, CO_2 is non-lubricating ("dry") so that the cups and centralizers of ILI tools and the seals of pumps and compressors must be selected to prevent premature wear due to elevated frictional effects in the non-lubricating CO_2 environment. For inline inspection tools, the length of the tool run is usually limited by the rate of wear on the cups and centralizers.

Public Awareness and Education

Compared with traditional gas pipelines, carbon dioxide pipelines are new, and present new risks to populations along their routes because, unlike natural gas, CO_2 is heavier than air which increases the risk of asphyxiation. Although the overall mitigated risk associated with carbon dioxide pipelines is lower than that of natural gas pipelines, carbon dioxide pipelines cause greater concern to stakeholders and affected communities along their length because concern over the new risk of asphyxiation. The solution has two parts – reduce the risks associated with carbon dioxide pipelines to as low as reasonably practicable (ALARP) and conduct more comprehensive community outreach programs to explain the safety features built into the pipeline and the actions to take in the highly unlikely event the carbon dioxide pipeline bursts, leaks, or is blown down.

Risk Profiles

Figure 8 illustrates conceptually the differences between the risk profiles of carbon dioxide pipelines and natural gas pipelines (Cooper and Barnett 2014). The two pipelines have different risk profiles because of differences in the properties of the two fluids. The major risk of carbon dioxide pipelines is asphyxiation because it is heavier than air and it can drift long distances from the point of release, while the major risk of natural gas pipelines is fire and explosion localized around the point of release.

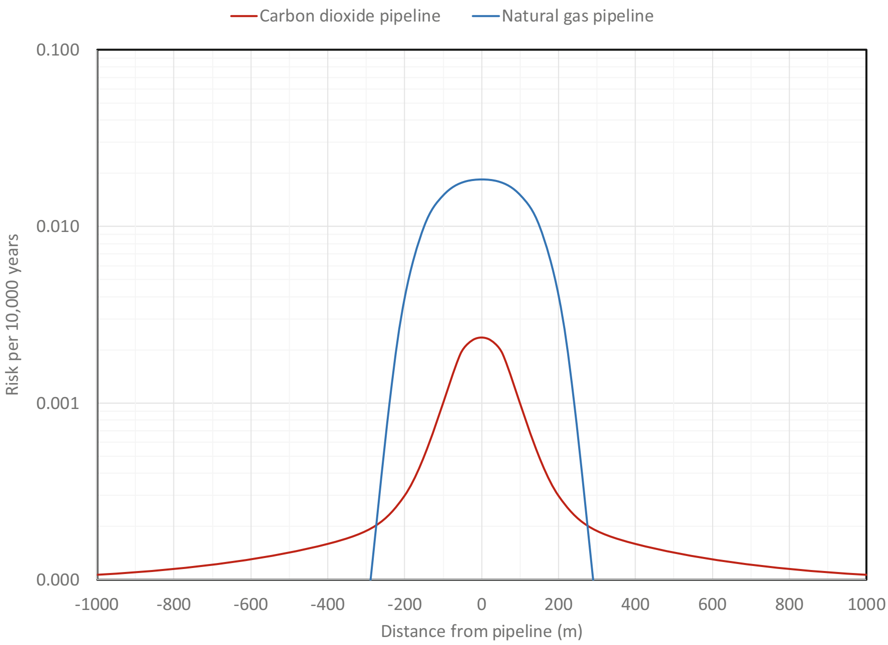

Fig. 8 Conceptual individual unmitigated risk profiles for natural gas and carbon dioxide pipelines (Cooper and Barnett 2014)

Both natural gas and carbon dioxide pipelines expose people and structures close to the burst site to severe physical consequences due to the operating pressure at the time of the burst. This means that bursts of dense phase pipelines can be expected to have higher physical consequences than bursts of natural gas pipelines, and bursts of gas phase carbon dioxide pipelines can be expected to have lower consequences than bursts of natural gas pipelines. But the fire and explosion risk of natural gas pipelines overshadows the mechanical risk associated operating pressure.

The net result is that within a 250 m wide corridor, carbon dioxide pipelines are safer than natural gas pipelines, but the risk diminishes more slowly with distance for carbon dioxide pipelines than for natural gas pipelines because the plume of CO_2 can drift with the wind or settle in low-lying areas and threaten communities remote from the pipeline right of way.

Emergency Response Plans

The main differences between CO_2 and natural gas that affect emergency response plans are that natural gas is flammable, while CO_2 is heavier than air. The important result of these two differences is that emergency response plans for CO_2 pipelines can be more complex than for natural gas pipelines because more time is available to respond to the asphyxiation threat of CO_2 pipelines than is available to respond to the fire and explosion threat of natural gas pipelines. This ability to mitigate the largest risk of CO_2 pipelines means that the mitigated overall risk of CO_2 pipelines can be made less than the mitigated overall risk of natural gas pipelines.

Features that can be built into the physical pipeline to minimize risks (such as minimizing the volume and frequency of releases, blowdown stack design, improved leak detection systems, and community-based CO_2 sensors) have been discussed in previous sections. A document that aids an owner-operator in the development of an emergency management program is CSA Z246.2 – *Emergency preparedness and response for petroleum and natural gas industry systems* (CSA Group 2018).

The priority of the emergency response plan is to ensure life safety, incident stabilization, and the protection of property and the environment. It is therefore important to identify the nature of an emergency and its location as quickly as possible. This process can be speeded up considerably using new technologies such as drones and fiber optic systems that can provide visibility and detail about the size and location of any release. A phone emergency notification system can be set up with cell phone operators along the pipeline route.

Residents and workers in communities in or near the path of the plume should be notified by an automated cell phone emergency alert backed up by an automatic computer driven call system with enough bandwidth to simultaneously alert all those at risk that a release of a given magnitude at a given location has been detected and with instructions about whether to evacuate or shelter in place. The automatic system should repeat-call unanswered calls at intervals and call alternative phone numbers if they are available. Emergency services such as fire, police, and ambulance must also be made aware; however, it is important they be trained ahead of time to understand their roles. To ensure adequate preparation for emergencies, mutual aid agreements are strongly recommended with communities, emergency services, and cell phone operators.

An Incident Commander, who has been trained appropriately and is qualified for the role, is responsible for ensuring the implementation of each subsequent step of the chosen emergency response plan and for upgrading or downgrading the level of response as necessary when more information is obtained. He is responsible for ensuring adequate pipeline isolation of the release zone (if applicable), such as closing sectionalizing valves (if they have not closed automatically on detection of rarefaction waves characteristic of large release events), shutting down pump/compressor stations, and for mobilizing the appropriate teams of first responders, which could include local fire rescue units, emergency medical services, and police. Detailed dispersion modelling can be incorporated into the emergency plan to predict the geographical limit of hazardous CO_2 levels, thereby allowing the extent of the disruption of the emergency response to be minimized.

The possibility of no-wind atmospheric conditions must be considered in the emergency response planning because sheltering in place can only be guaranteed effective for a few hours. If there is no wind and the cloud of CO_2 is stationary and does not disperse with time, the Incident Commander might decide to issue evacuation orders. The evacuation plan should consider the use of electric motor driven buses equipped with compressed air tanks and SCBA gear, garaged at locations along the length of the pipeline selected to keep travel times to low-lying terrain areas along the route within acceptable limits. The buses can then be manned by first responders to rescue people as needed and evacuate them to safe locations outside the geographical boundaries of hazardous CO_2 levels.

The emergency response plans should be reviewed and revised annually, and all technical participants and first responders should receive annual training. The phone number list should be updated annually.

Community Outreach

Because the success of the emergency response plan depends on community self-help, community information sessions are essential, particularly regarding the decision to either evacuate or shelter in place. These community information sessions should begin early in the permitting process to emphasize the safety features built into the pipeline and the high level of safety of carbon dioxide pipelines to allay fears and objections to the pipeline. The community information sessions should then continue annually throughout the life of the pipeline to assure community readiness, the viability of the shelter-in-place option and what to expect if evacuation is necessary, so that fatalities can be prevented in the unlikely event of a large release of CO_2.

Conclusion

Design and operation of carbon dioxide pipelines is different from natural gas pipelines because the properties of CO_2 are different from natural gas. In areas where design and operation are the same, the pipeline industry has extensive real-world experience that can be relied upon to develop safe cost-effective solutions. But in areas where they are different, pipeline designers and operators must rely on fundamental engineering principles. The areas in which the design and operation of carbon dioxide pipelines differ from natural gas pipelines have been identified and recommended best practices for dealing with the differences have been presented. The solutions involve changes to many details including operating conditions, material selection, compositional limits, operating procedures, and community relations. If the recommendations are followed, safe and economical transportation of CO_2 by pipeline can be achieved.

References

American Society of Mechanical Engineers (2022) ASME B31.4 - 2022 pipeline transportation systems for liquids and slurries. ASME

Botros GRR (2017) Measurements of decompression wave speed in binary mixtures of carbon dioxide and impurities. J Pressure Vessel Technol 139

CIA Factbook (2015). https://en.wikipedia.org/wiki/List_of_countries_by_total_length_of_pipelines. (Accessed 10 Dec 2022)

Cooper R, Barnett J (2014) Pipelines for transporting CO_2 in the UK. Energy Procedia 63: 2412–2431
Cosham A, Hopkins P, Jones DJ, Barnett J (2013) Warm pre-stressing and its relevance to CO_2 pipelines, Paper S05-02. Pipeline Technology, Oostende
CSA Group (2018) CSA Z246.2 Emergency preparedness and response for petroleum and natural gas industry systems. CSA Group
CSA Group (2019) CSA Z662–19 oil and gas pipelines. CSA Group, Toronto
DNV GL (2019) DNV-RP-F104 design and operation of carbon dioxide pipelines. DNV GL
Doctor R, Palmer A, Coleman D, Hendriks JDC, Kaarstad O, Ozaki M (2005) IPCC special report on carbon dioxide capture and storage
Global CCS Institute (2022) Global status of CCS 2022
Han MS (1972) A generalized correlation for prediction of fluid thermodynamic properties and phase behavior and its industrial applications. University of Oklahoma, Norman
IEA (2022) CO_2 transport and storage
IGI Global (2022) What is saturation line [Online]. Available: https://www.igi-global.com/dictionary/high-power-heat-transfer-in-supercritical-fluids/89494. (Accessed 27 Dec 2022)
King G (1979) Decompression of gas pipelines during longitudinal ductile fractures. J Energy Resour Technol 101
G. G. King and D. L. Katz, "Dense Phase Transmission of Natural Gas.," *Energy Processing Canada,* 1973
Maxey WA, Kiefner JF, Eiber RJ (1976) Ductile fracture arrest in gas pipelines
Morland BH, Tjelta M, Dugstad A, Svenningsen G (2019) Corrosion in CO_2 systems with impurities creating strong acids. Corrosion, NACE
Morland BH, Dugstad A, Svenningsen G (2021) Experimental based CO_2 transport specification ensuring material integrity. In: 15th international conference on greenhouse gas control technologies, GHGT-15
Peletiri SP, Rahmanian N, Mujtaba IM (2018) CO_2 pipeline design: a review. Energies 2184:25
Pistone V, Re G, Vogt G, Demfonti G, Jones D (1993) EPRG recommendations for crack arrest toughness for high strength pipe steels. In: Proceedings of the Eight symposium online pipe research
Span R, Wagner W (1996) A new equation of state for carbon dioxide covering the fluid region from the triple-point temperature to 1100 K at pressures up to 800 MPa. J Phys Chem Ref Data 25(6)
Starling K, Powers J (1970) Enthalpy of mixtures by modified BWR equation. Ind Eng Chem Fundam 9:531–537
Svenningsen G, Morland BH (2021) Corrosion of carbon steel in simulated CCS streams. In: 15th international conference on greenhouse gas control technologies, GHGT-15
US Department of Transportation, Pipeline and Hazardous Materials Safety Administration (2020) Failure investigation report, pipeline rupture/natural force damage
Varzandeh SY (2017) Comparison of GERG-2008 and simpler EOS models. Fluid Phase Equilib 434:21–43
Wang H, Chen J, Li Q (2019) A review of pipeline transportation technology of carbon dioxide. In: IOP conference series: earth and environmental science, vol 310
Wikipedia (2017). 13 Mar 2017. Available: https://en.wikipedia.org/wiki/Bubble_point. (Accessed 27 Dec 2022)
Wikipedia (2022). 10 Sept 2022. Available: https://en.wikipedia.org/wiki/Hydrocarbon_dew_point. (Accessed 27 Dec 2022)
Zhao Q, Mecheri M, Neveux T, Privat R, Jaubert J-N (2017) Selection of a proper equation of state for the modeling of a supercritical CO_2 Brayton cycle: consequences on the process design. Ind Eng Chem Res 56(23):6841–6853